

武汉大学数学与统计学院 2025-2026 第二学期

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Each ring in this exam is commutative with a unit

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- (1) State the three equivalent characterizations of Noetherian rings.
- (2) State the Hilbert Nullstellensatz.
- (3) State the going-up and going-down theorem.

2、 Let $A = A_1 \times A_2$.

- (1) Prove that every ideal I of A is of the form $I_1 \times I_2$, where I_1, I_2 are ideals.
- (2) If A_1, A_2 are principal ideal rings, prove that A is principal.

3、

- (1) Let $\mathfrak{a} \subset A$ be an ideal, and $f : A \rightarrow B$ be a ring homomorphism. Suppose that B is a flat A -algebra. Prove that the natural homomorphism $\mathfrak{a} \otimes_A B \rightarrow \mathfrak{a}B$ is an isomorphism.
- (2) Let $\mathfrak{a}_1, \mathfrak{a}_2$ be ideals of A . Prove that

$$(\mathfrak{a}_1 \cap \mathfrak{a}_2)B = (\mathfrak{a}_1B) \cap (\mathfrak{a}_2B).$$

4、 Let $A \subset B$, with B integral over A .

- (1) Prove that $A^\times = B^\times \cap A$.
- (2) Prove that $\text{Rad}(A) = \text{Rad}(B) \cap A$.

5、 Prove the Five Lemma.

6、 Let A be Noetherian, $\phi : M \rightarrow N$ be a finitely-generated module homomorphism, and $\mathfrak{p} \in \text{Spec}(A)$.

- (1) If $\phi_{\mathfrak{p}}$ is injective, prove that $\exists f \in A - \mathfrak{p}$ such that $\phi_f : M_f \rightarrow N_f$ is injective.

(2) If $\phi_{\mathfrak{p}}$ is surjective, prove that $\exists g \in A - \mathfrak{p}$ such that ϕ_g is surjective.

7、 Let $\mathfrak{m} \in \text{MaxSpec}(A)$.

(1) Prove that: $A/\mathfrak{m}^n \cong A_{\mathfrak{m}}/(\mathfrak{m}_{\mathfrak{m}})^n$.

(2) Use Problem 2 and 7(1) to prove that if A is a Dedekind domain, then every ideal of A is generated by at most two elements.